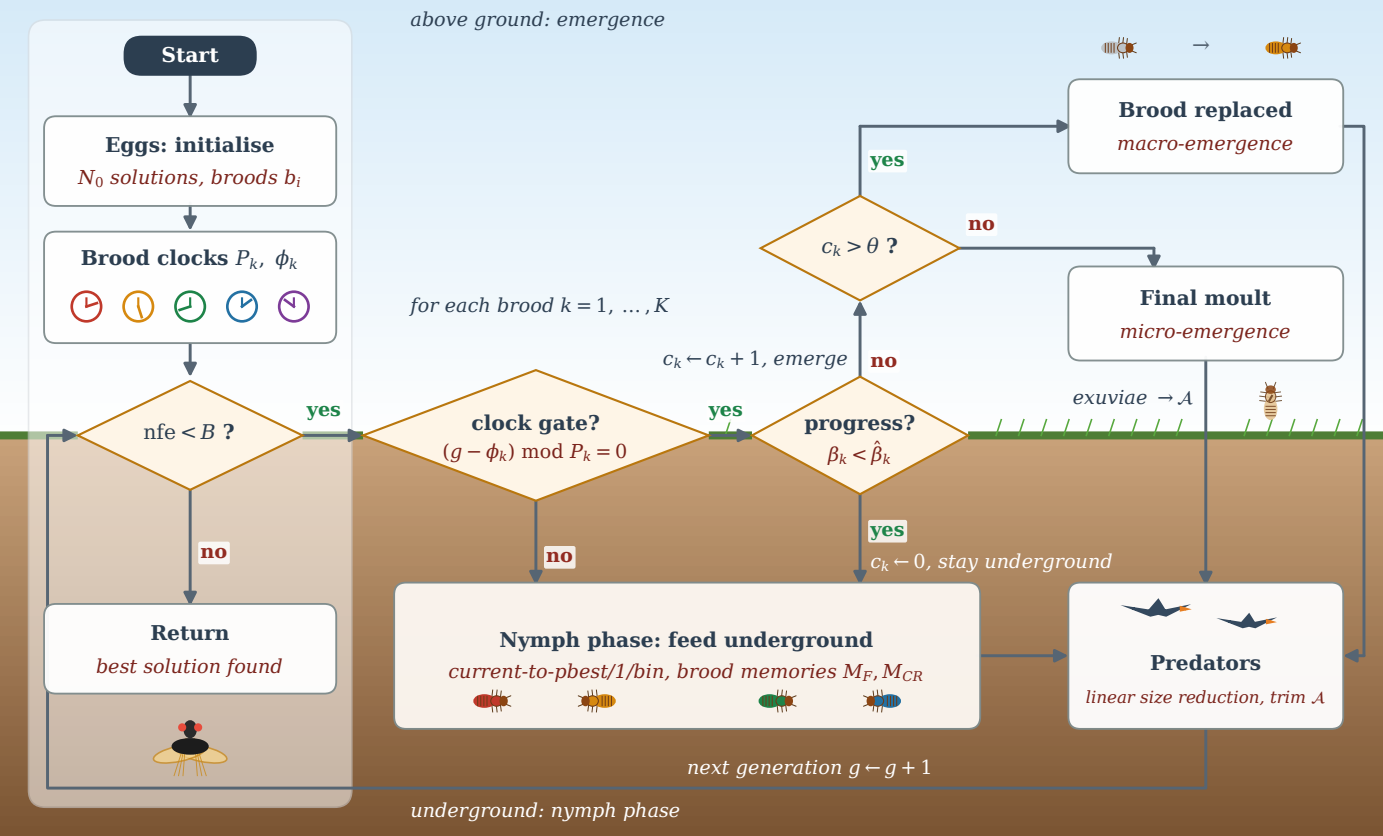
above ground: emergence
Start
Eggs: initialise
N_0 solutions, broods b_i
Brood clocks P_k, ϕ_k
nfe < B ?
yes
no
Return
best solution found
for each brood $k = 1, \ldots, K$
clock gate?
$(g - \phi_k) \bmod P_k = 0$
yes
no
progress?
$\beta_k < \hat{\beta}_k$
yes
$c_k \leftarrow 0$, stay underground
$c_k > \theta$?
yes
no
$c_k \leftarrow c_k + 1$, emerge
Brood replaced
macro-emergence
Final moult
micro-emergence
exuviae $\rightarrow \mathcal{A}$
Nymph phase: feed underground
current-to-pbest/1/bin, brood memories M_F, M_{CR}
Predators
linear size reduction, trim $\mathcal{A}$
next generation $g \leftarrow g + 1$
underground: nymph phase